

TECHNICAL ASSESSMENT

Question Answering System: Design Brief

For the role of NLP Engineer

ISSUED BY	Verbalis AI, Language Systems Group
ROLE	NLP Engineer (Product Engineering)
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	30 June 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

In today's digital world, organizations manage vast amounts of unstructured text such as knowledge bases, legal documents, product manuals, healthcare guidelines, and customer support articles. Finding precise information from these documents manually is time-consuming and inefficient. This project addresses that challenge by developing an Extractive Question Answering System that can identify and return the exact answer to a user's question directly from the provided context using transformer-based NLP models.

2. PROBLEM STATEMENT

Design and implement a system that takes a user question and a passage of text, and returns the answer to the question. The system shall use pretrained transformer language models rather than rule-based or keyword matching methods.

3. OBJECTIVE

Given a question and one or more context, the system shall return the answer, or the most likely answers, drawn from the context. The answers shall be ranked, with the most likely answer first. The system shall use transformer-based language models to represent and process text.

4. FUNCTIONAL REQUIREMENTS

4.1 Text Preprocessing

The system shall accept a question and a context passage as plain text, and apply the same tokenisation steps to both, using a Hugging Face tokenizer suited to the chosen transformer model.

4.2 Text Representation

The system shall represent the question and context using contextual embeddings produced by a pretrained transformer model, loaded through the Hugging Face Transformers library.

4.3 Model Inference and Fine-Tuning

The system shall use a Hugging Face inference pipeline to answer questions out of the box, and shall support fine-tuning of the model on a custom dataset using PyTorch and the Hugging Face Datasets library.

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4.4 Answer Extraction and Confidence Scoring

The system shall extract the answer span from the context, assign a confidence score to each candidate answer, and return answers in ranked order, most confident first.

4.5 Model Saving and Reuse

The system shall let the user save a fine-tuned model to disk and reload it for later use without retraining, and shall let the user set how many candidate answers are returned for each question.

4.6 Generalisation

The system shall allow the fine-tuned model to be saved and reloaded without requiring retraining. The saved model shall be integrated into a Flask-based web application for real-time question answering. Users shall be able to specify the number of candidate answers returned for each query.

5. INPUT AND OUTPUT SPECIFICATION

Input: A user question in plain text, and one or more context passages .

Output: A ranked list of candidate answers, each with a confidence score, with the most likely answer first.

6. EVALUATION CRITERIA

6.1 Answer Quality	The returned answers are correct and relevant to the question, and the most likely answer appears first.
6.2 Robustness	The system handles different question types, messy text, and long or noisy context passages.
6.3 Generalisation	The system works on questions and documents that were not used during development.
6.4 Clarity of Approach	The candidate clearly explains the method and the reasons for the main choices.
6.5 Code Quality	The code is clear, organised, and easy to run.

7. SUBMISSION REQUIREMENTS

Candidates shall submit the following:

- Source code
- Report
- README
- Output screenshots
- Short explanation video

The explanation video is required. It shall cover: (a) the overall approach and the main choices; (b) how the system works, from preprocessing to answer ranking; and (c) a short run through one question, showing how the answer is produced.

A submission without the video will be treated as incomplete.

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8. TIMELINE

The assessment shall be completed and submitted within 48 hours from the time of issue. Late submissions will not be considered unless an extension is agreed in writing in advance.

Note to candidate: You may use any public text dataset and any common Python libraries (for example Hugging Face Transformers, Datasets, and PyTorch), as long as you list them in your explanation. If anything is unclear, please ask before you start.